Multimode Raman Baselines and Cost Choice

Fiber Raman Suppression Project

Evidence snapshot: 2026-04-28

Abstract

This note summarizes the current multimode-fiber Raman-suppression lane. The accepted result is a qualified simulation finding: in an idealized six-mode GRIN-50 model, one shared spectral phase can reduce the simulated Raman-band fraction by 31.7 dB while passing corrected temporal-edge diagnostics. The most important part of the story is that an earlier large-gain solution was rejected after raw edge diagnostics exposed temporal-window contamination. The current result is therefore meaningful, but not a generic experimental claim about arbitrary multimode fibers.

Claim Status

Status: qualified simulation result
The MMF lane is closed/exploring. It is usable as a presentation-ready simulation result with clear claim boundaries: shared spectral phase shaping suppresses Raman-band generation in one idealized GRIN-50 setting under strict diagnostics. It is not yet a paper-grade or experiment-grade universal MMF result.

1 Question and Thesis

The single-mode notes optimize one spectral phase for one propagation mode. In a multimode fiber, the field has several mode amplitudes:

$$\hat{u}_{m,k}(z),$$

where m indexes mode and k indexes frequency. The question is:

Can one shared spectral phase reduce Raman generation across a multimode field?

The current evidence says yes in a constrained simulation. The accepted case is GRIN-50 with six scalar modes, length $L = 2\,\mathrm{m}$, power $P = 0.20\,\mathrm{W}$, $N_t = 4096$, and a 96 ps time window. The optimized shared phase reduces the summed Raman-band objective from $-17.96\,\mathrm{dB}$ to $-49.69\,\mathrm{dB}$.

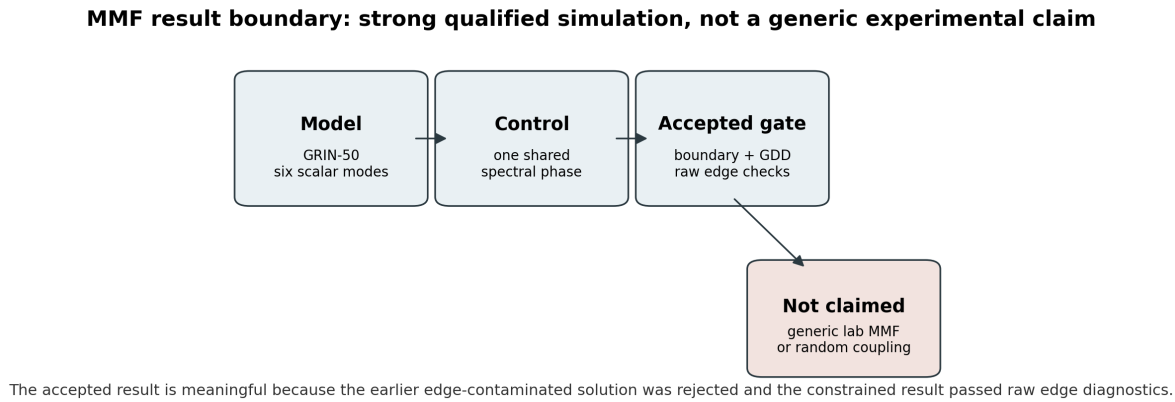


Figure 1: The current MMF result is intentionally narrow. The project has a credible constrained simulation candidate, not a generic experimental claim for all multimode fibers or all launch conditions.

2 Why MMF Is Subtle

Multimode fibers are not just single-mode fibers with a bigger vector. Raman energy can appear differently by mode, launch composition matters, and mode coupling can change the nonlinear evolution. Experiments and simulations of graded-index multimode fibers show mode-resolved Raman cascades [1], nonlinear beam self-cleaning [2], and launch/wavefront sensitivity [3, 4].

That literature shapes the claim boundary. A summed output spectrum is not enough. A credible MMF result needs:

- a clear objective: summed, fundamental-mode, or worst-mode Raman fraction;
- mode-resolved inspection, not only a total spectrum;
- temporal-window diagnostics, because pseudospectral propagation uses periodic time windows [5];
- an explicit statement of launch and coupling assumptions.

TL;DR

The MMF result is harder to explain because there are two kinds of structure: frequency structure and mode structure. The shared phase only controls the frequency structure. It does not independently control the transverse modal wavefront, and it does not prove robustness to every possible launch.

3 Model, Objective, and Gradients

The accepted model is:

$$\text{GRIN-50}, \quad M = 6, \quad L = 2\,\text{m}, \quad P = 0.20\,\text{W}.$$

The same spectral phase is applied to all modes:

$$\hat{u}_{m,k}(0;\phi) = \hat{u}_{m,k,0} e^{i\phi_k}.$$

The primary objective is the mode-summed Raman fraction:

$$J_{\text{sum}} = \frac{\sum_m \sum_{k \in \mathcal{B}_{\text{Raman}}} |\hat{u}_{m,k}(L)|^2}{\sum_m \sum_k |\hat{u}_{m,k}(L)|^2}.$$

The diagnostic report also computes a fundamental-mode fraction and worst-mode fraction so the improvement cannot hide inside a summed detector. For the accepted run:

$$J_{\text{sum}} = -49.69\,\text{dB}, \qquad J_{\text{fund}} = -49.65\,\text{dB}, \qquad J_{\text{worst}} = -45.35\,\text{dB}.$$

The shared-phase gradient sums the modal sensitivities:

$$\frac{\partial J}{\partial \phi_k} = 2\,\text{Re} \sum_m \lambda_{m,k}(0)^*\, i\, \hat{u}_{m,k}(0).$$

This is the same phase-chain rule as the single-mode case, except every mode contributes to the same phase pixel.

Accepted Regularized Surface

The accepted constrained run uses a dB-scaled surface of the form

$$10 \log_{10} \left(J_{\text{sum}} + \lambda_{\text{gdd}} R_{\text{gdd}} + \lambda_{\text{boundary}} R_{\text{boundary}} \right),$$

with

$$\lambda_{\text{gdd}} = 10^{-4}, \qquad \lambda_{\text{boundary}} = 0.05.$$

The boundary term is central to this note because it distinguishes the rejected result from the accepted one.

4 Diagnostic Correction

The earlier MMF result looked strong, but it was not accepted. The problem was temporal-window trust. The corrected convention is:

$$\hat{u} = \text{ifft}(u_t), \qquad u_t = \text{fft}(\hat{u}).$$

After this correction, the unregularized optimized output still had about 5.02×10^{-2} temporal-edge energy. That is too high, so the result was rejected as a boundary artifact.

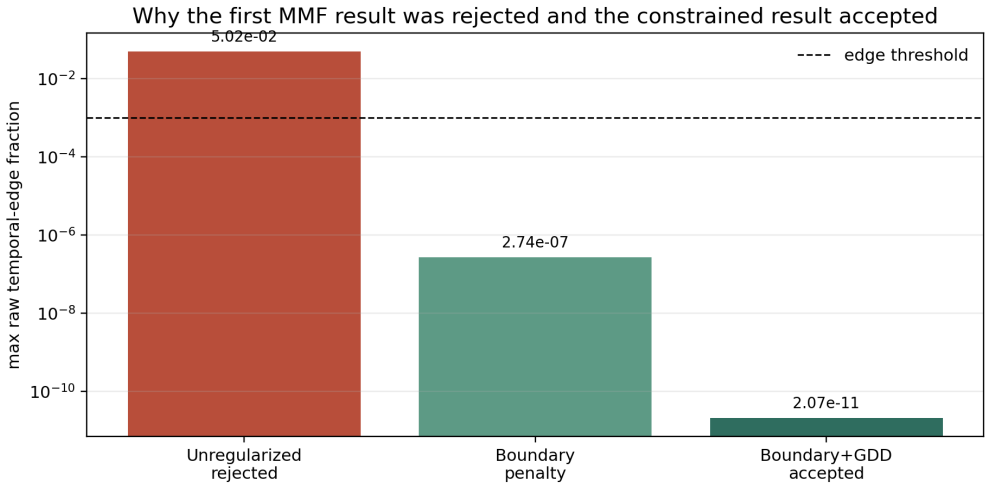


Figure 2: The unregularized solution is rejected because its raw edge fraction is too high. Boundary and GDD regularization recover strong suppression while driving the raw edge fraction far below the trust threshold.

TL;DR for the Trust Gate

If a pulse reaches the edge of the simulated time window, the FFT can wrap it around as if time were periodic. A result can then look good for the wrong reason. The accepted MMF result is important because it survives the raw-edge check, not just because it has a low Raman number.

5 Implementation Surface

Component	Role
MMF setup	Builds the GRIN-50 preset, mode launch, grid, fiber, and Raman band.
Shared-phase optimizer	Computes the MMF forward solve, adjoint solve, summed objective, and shared phase gradient.
Window-validation driver	Reruns the threshold MMF case with larger windows and regularizers, then writes the validation summary.
Cost helpers	Provide summed, fundamental-mode, and worst-mode Raman fractions.
Standard-image helper	Saves phase profile, phase diagnostic, optimized heat map, and no-shaping heat map.
Front-layer config	Supports planning and dry-run validation only; real MMF propagation remains burst/heavy-compute work.

6 Experimental Ladder

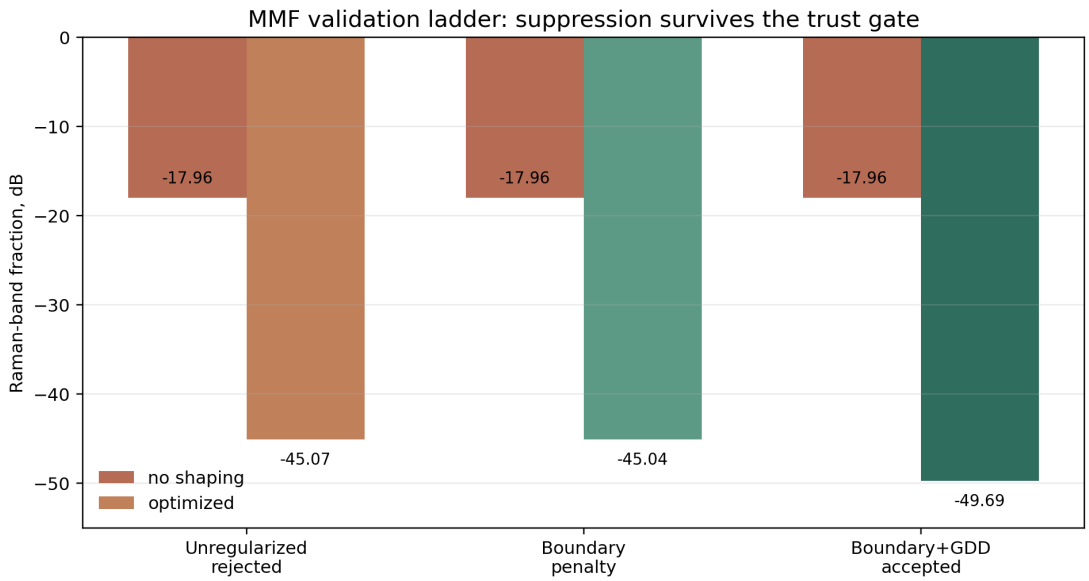


Figure 3: MMF validation ladder. The final accepted result is not just the deepest-looking run; it is the deepest run that also passes the raw temporal edge diagnostics among the tested candidates.

Case	**Objective dB**	**Improvement**	**Max edge**	**Verdict**
Unregularized	-45.07	$27.12\,\mathrm{dB}$	5.02×10^{-2}	reject
Boundary penalty	-45.04	$27.09\,\mathrm{dB}$	2.74×10^{-7}	pass
Boundary+GDD	-49.69	$31.73\,\mathrm{dB}$	2.07×10^{-11}	accepted

Table 1: All rows use $L = 2\,\mathrm{m}$, $P = 0.20\,\mathrm{W}$, $N_t = 4096$, and $96\,\mathrm{ps}$. The unregularized row is scientifically useful because it explains what had to be fixed, but it is not accepted as a valid result.

7 Representative Result Pages

7.1 No-Shaping Control

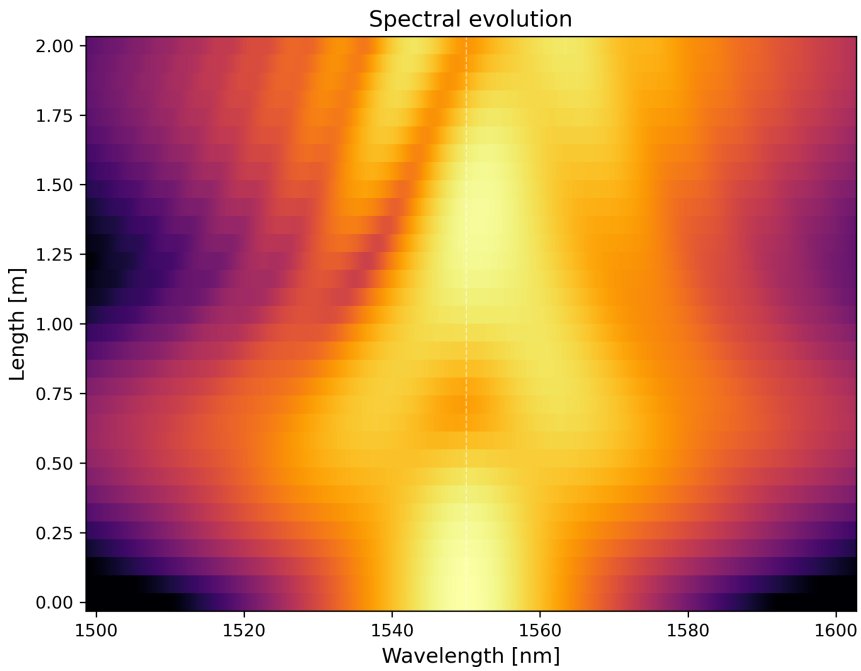
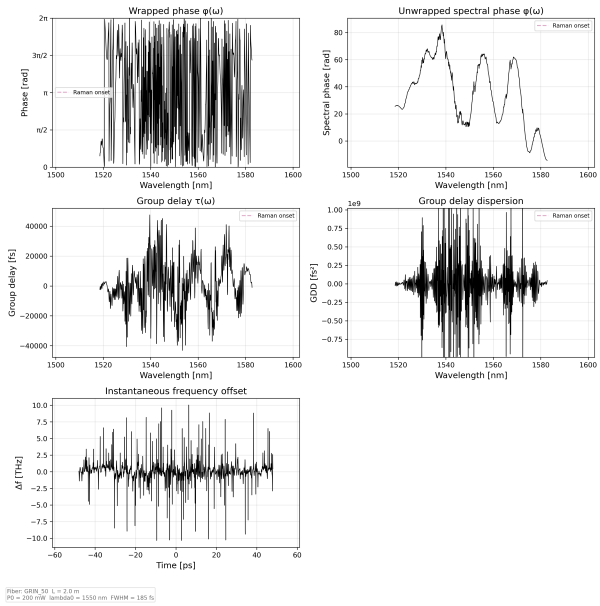
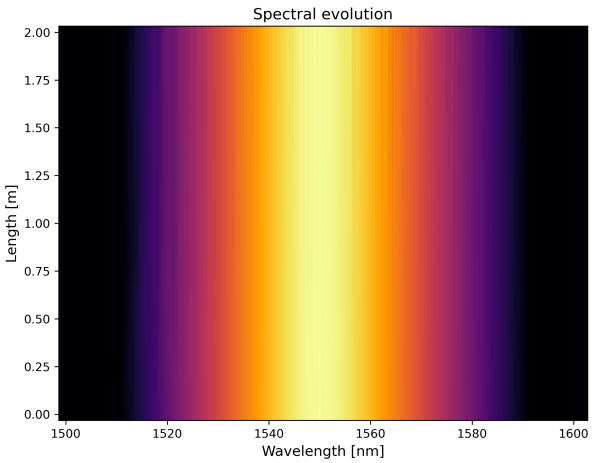


Figure 4: No-shaping control for the accepted MMF operating point. This is the baseline heat map for judging whether the shared spectral phase changed Raman growth.

7.2 Rejected Unregularized Run



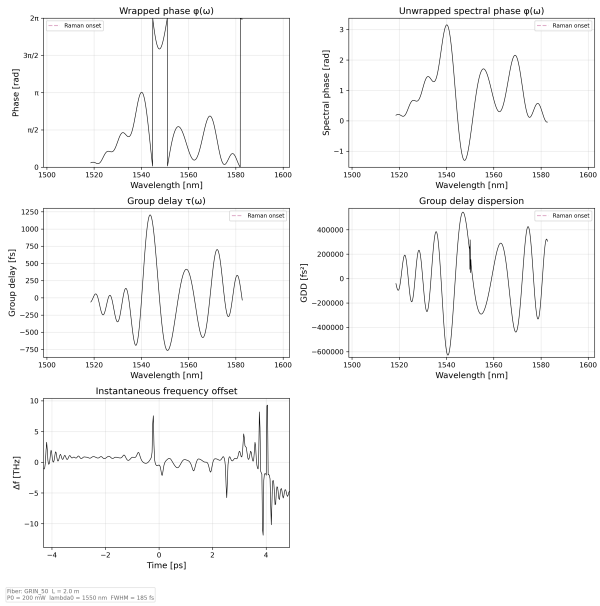
Rejected unregularized phase diagnostic.



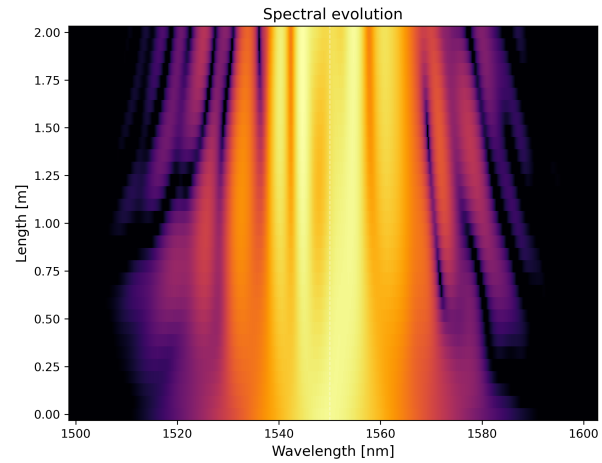
Rejected unregularized heat map.

Figure 5: The unregularized run visually suppresses Raman output, but it is not accepted because corrected raw-edge diagnostics show temporal-window contamination.

7.3 Accepted Boundary+GDD Run

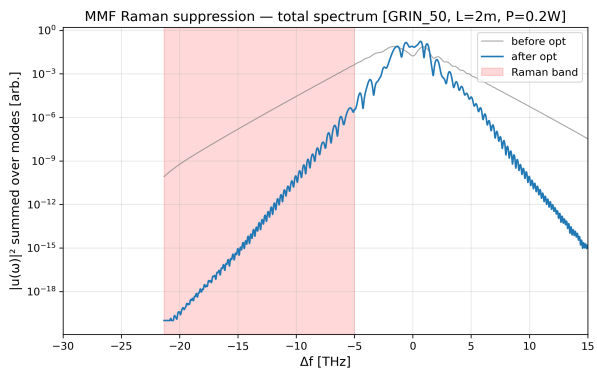


Accepted phase diagnostic.

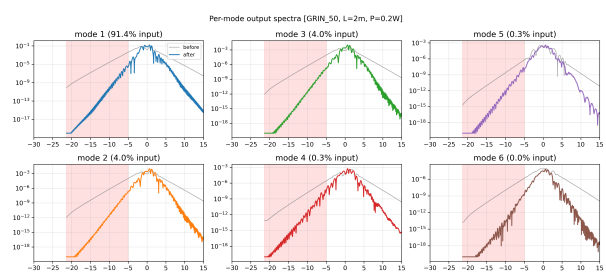


Accepted optimized heat map.

Figure 6: Accepted constrained MMF result: $-17.96 \rightarrow -49.69\,\mathrm{dB}$ under the mode-summed Raman objective, with raw edge fractions near 2×10^{-11}.

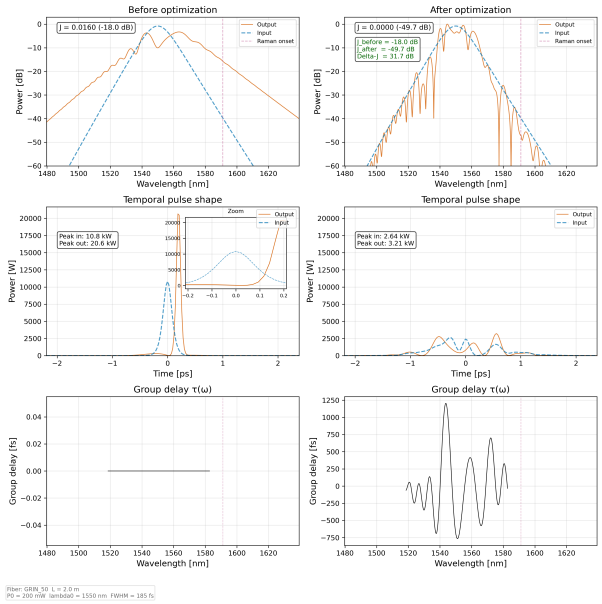


Total spectrum.

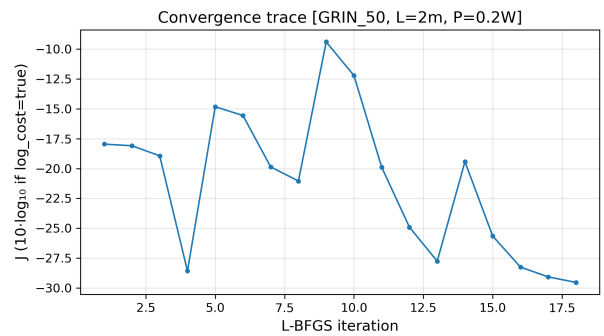


Per-mode spectrum.

Figure 7: Mode-resolved inspection matters. The accepted run is not only a summed-detector accounting win; the launched modes also show suppression.



Accepted phase profile.



Convergence trace.

Figure 8: The accepted phase is still engineered and oscillatory. This is a simulation candidate, not a lab actuator prescription.

8 Interpretation

The MMF lane changed from "blocked by invalid-window" to "qualified simulation result." The key scientific behavior survives after the trust correction: boundary and GDD penalties can keep the temporal window clean while still producing large Raman suppression.

The result should be presented with discipline. It supports shared spectral phase as a possible control mechanism in an idealized GRIN-50 MMF model. It does not yet prove robustness to launch changes, random mode coupling, more modes, experimental wavefront shapers, or a larger $N_t = 8192$ grid.

9 Presentation Capsule

If this note becomes a presentation block, the clean message is:

- MMF is harder because Raman is mode-resolved and launch/coupling assumptions matter.
- The first large-gain result was rejected because the pulse touched the temporal window boundary.
- The accepted constrained run still gives a 31.7 dB reduction and passes raw edge diagnostics.
- The honest claim is an idealized six-mode GRIN-50 simulation result, not generic experimental MMF Raman suppression.

10 Limitations

- **No broad experimental claim.** The model is idealized and uses six scalar modes.
- **Shared spectral phase only.** This is not independent spatial wavefront control of each mode.
- **Launch sensitivity open.** Default launch is LP01-heavy; other launch compositions still need a matrix of checks.
- **Random coupling open.** The accepted result does not include a random or degenerate mode-coupling robustness sweep.
- **Grid-refinement gate open.** The $N_t = 8192$, 96 ps refinement is currently blocked by memory/compute availability.
- **Phase realism open.** The accepted phase has sub-ps to roughly ps-scale group-delay structure and should be simplified before lab claims.

11 Reproduction Capsule

Planning-only front-layer check:

```
julia -t auto --project=. scripts/canonical/run_experiment.jl \
  --dry-run grin50_mmf_phase_sum_poc
```

Accepted constrained candidate, to run on burst or equivalent heavy compute:

```
MMF_VALIDATION_SAVE_DIR=results/raman/mmf_validation_gdd \
MMF_VALIDATION_CASES=threshold \
MMF_VALIDATION_MAX_ITER=4 \
MMF_VALIDATION_THRESHOLD_TW=96 \
MMF_VALIDATION_THRESHOLD_NT=4096 \
MMF_VALIDATION_LAMBDA_BOUNDARY=0.05 \
MMF_VALIDATION_LAMBDA_GDD=1e-4 \
julia -t auto --project=. scripts/research/mmf/mmf_window_validation.jl
```

Expected closeout:

- validation summary table;
- phase profile and phase diagnostic;
- optimized and no-shaping heat maps;
- total and per-mode spectra;
- raw input/output temporal-edge fractions;
- explicit statement of whether the result is accepted, rejected, or compute-blocked.

12 Sources

References

[1] H. Pourbeyram, G. P. Agrawal, and A. Mafi. "Stimulated Raman scattering cascade spanning the wavelength range of 523 to 1750 nm using a graded-index multimode optical fiber." https://arxiv.org/abs/1301.6203

[2] K. Krupa et al. "Spatial beam self-cleaning in multimode fibres." *Nature Photonics*, 2017. https://doi.org/10.1038/nphoton.2017.32

[3] E. Deliancourt et al. "Wavefront shaping for optimized many-mode Kerr beam self-cleaning in graded-index multimode fiber." https://arxiv.org/abs/1902.04453

[4] O. Sidelnikov et al. "Effect of random mode coupling on Kerr beam self-cleaning in graded-index multimode fibers." https://arxiv.org/abs/1908.07745

[5] O. Melchert and A. Demircan. "A python package for ultrashort optical pulse propagation in terms of forward models for the analytic signal." https://arxiv.org/abs/2104.11649